EXP NO : 01
DATE : 23.07.26

STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

Aim :
To study the various network commands used in linux and windows.

Basic networking commands

1. arp-a : ARP is the short form of address resolution protocol, It will show the IP address of your computer along with the IP address and MAC address of your router.

Output :

Interface : 172.16.10.67 --- 0xd

Internet Address	Physical Address	Type
172.16.8.1	7c-5a-1c-cf-be-45	dynamic
172.16.8.9	d4-3d-7e-ad-40-6a	dynamic
172.16.8.10	f8-bc-12-8f-cd-7a	dynamic.

2. Hostname : This is the simplest of all TCP/IP commands. It displays the name of your computer.

Output :
DESKTOP-D1G1OCPN

3. ipconfig/all : This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.

Output :

Windows IP Configuration

Hostname ..... : DESKTOP-D1G1OCPN
Primary DNS Suffix ..... :

Node Type . . . . . : Hybrid
IP Routing Enabled . . . . : No
WINS Proxy Enabled . . . . . : No

4. nbtstat -a : This command helps solve problems with NetBIOS name resolution (NBt stands for NetBIOS over TCP/IP)

Output:

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [Interval]]

-a (adapter status) Lists the remote machine's name table given its name
-A (adapter status) Lists the remote machine's name table given its IP address.
-c (cache) Lists NBT's cache of remote [machine] names and their IP addresses.

5. netstat : (network statistics) netstat displays a very variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incomming and outgoing as well as viewing routing tables, interface statistics etc.

Output:

Active connections

Proto	Local address	Foreign Address	State
TCP	172.16.10.67:55991	4:213.25.241:https	ESTABLISHED
TCP	172.16.10.67:56207	s1-in-f188:5228	ESTABLISHED
TCP	172.16.10.67:56375	18.107.213.254:https	CLOSE-WAIT

6. nslookup : (name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in two modes : interactive and non-interactive.

output:
Default server : Unknown
Address : 172.16.8.1

7. pathping : pathping is unique to windows and is basically a combination of the ping and Tracert commands : pathping traces the route to the destination address the launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

Output:

usage : pathping [-g host-list] [-h maximum_hops] [-i address] [-n] [-p period] [-q num_queries] [-w timeout] [-4] [-6] target_name.

options:
-g host-list — Loose source route along host-list
-h maximum_hops — maximum number of hops to search for target

8. ping : (packet Internet Groper) command is the best way to test connectivity between two nodes. Ping use ICMP to communicate to other devices.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name (ping www.facebook.com).

output:

usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL]
[-v TOS] [-r count] [-s count] [-j host-list] |
[-k host-list]] [-w timeout] [-R] [-S srcaddr]
[-c compartment] [-p] [-4] [-6] target_name

options:

-t Ping the specified host until stopped.
To see statistics and continue - type control-break
To stop - type control-C.

-a Resolve addresses to hostnames.

9. Route: route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

output:

Manipulate network routing table.

ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METRIC metric]
[IF interface]

-f clears the routing tables of all gateway entries. If this is used in conjunction with one of the commands, the tables are cleared prior to running the command.

10. ip address show: Displays all network interfaces and their assigned IP addresses.

output: enp0s3 UP

11. ip address add 192.168.1.254/24 dev enp0s3

Assigns the IP address 192.168.1.254 to interface enp0s3.

output:

12. ip address del 192.168.1.254/24 dev enp503
Removes the specified IP address from the interface
output:

13. ip link set eth0 up
Enables (activates) the network interface eth0.
output:

14. ip link set eth0 down
Disables (deactives) the network interface eth0.
output:

15. ip link set eth0 promisc on
Enables promiscous mode so the interface captures all network packets.
output:

16. ip route add ~~default~~ via 192.168.1.254
Adds a route to the 192.168.1.0/24 network through the specified gateway
output:

17. ip route add default via 192.168.1.254 dev eth0
Adds a default gateway for all outgoing network traffic
output:

18. ip route add 192.168.1.0/24 dev eth0
Adds a direct route to the specified route from the outgoing table
output:

2. ifconfig : The ifconfig command was/is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed above.

output:

eth0 : flags = 4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500
inet 192.168.1.100. netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::a00:2788:fe4e:66a1 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:4e:66:a1 txqueuelen 1000 (Ethernet)

3. mtr: MTR is a progress with command line interface that serves as a network diagnostic and trouble shooting tool.

output: My traceroute [v0.87]

Host	Packets					
	Loss %	Snt	Last	Avg	Best	Wrst	St dev
1. ::1	0.0 %	145	0.2	0.1	0.1	0.6	0.0

4. tcpdump:

The tcp dump command is designed for capturing and displaying packet.

output:
tcp dump : you don't have permission to capture on that device.

5. ping : Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending internet control message

output:
PING google.com (142.200.182.78) 56(84) bytes of data.
64 bytes from lemma-ane-in = 1 ttl = 12 time = 988 ms

Result:

Thus the basic networking as Linux commands are learned and the O/P get verified.